



P.E.S COLLEGE OF ENGINEERING MANDYA-571 401

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



**A Project Report
on**

“DEEP LEARNING-BASED TUBERCULOSIS DETECTION USING CHEST RADIOGRAPHS”

Submitted in partial fulfillment for the award of Degree of

**BACHELOR OF ENGINEERING
IN
INFORMATION SCIENCE AND ENGINEERING**

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**DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING
2025-2026**



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CERTIFICATE

This is to certify that **ARUN KUMAR 4PS22IS007, DEEKSHITH A 4PS22IS015, DHANUSH AM 4PS22IS017, VISHWABANDHU KN 4PS22IS059** has been successfully completed the Project work "**DEEP LEARNING-BASED TUBERCULOSIS DETECTION USING CHEST RADIOGRAPHS**" in partial fulfillment for the award of Bachelor of Engineering of PES College of Engineering, Mandya (An Autonomous Institution, Affiliated to VTU, Belagavi) during the year 2025-26. It is certified that all corrections indicated in internal assessment have been incorporated in the report deposited in the library. The Project Work have been approved as it satisfies the academic requirements in respect of work prescribed for the degree in Bachelor of Engineering in Information Science and Engineering discipline.

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DECLARATION

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ABSTRACT

Tuberculosis (TB) is a highly infectious disease caused by *Mycobacterium tuberculosis* and remains one of the leading causes of infectious disease mortality worldwide. Accurate and early detection of TB is critical for effective treatment and prevention of community transmission. Chest X-ray imaging is the most widely used and accessible diagnostic tool for TB screening, particularly in low-resource healthcare environments where specialist radiologists are scarce. However, manual interpretation of chest radiographs is time-consuming, subjective, and prone to inconsistency.

This project proposes an automated, deep learning-based TB detection system that classifies chest X-ray images as TB-positive or normal without requiring complex lung segmentation preprocessing. The system employs transfer learning on two backbone architectures — DenseNet121 and ResNet50 — initialized with ImageNet pre-trained weights and fine-tuned on a curated subset of approximately 4,200 images from the NIH Clinical Center Chest X-ray dataset. To address the class imbalance problem inherent in TB datasets, Generative Adversarial Networks (GANs) are used to synthesize high-quality TB-positive chest X-ray images, enriching the training data with diverse pathological patterns. Three training configurations are evaluated: DenseNet121 with standard augmentation, DenseNet121 with GAN-based augmentation, and ResNet50 with standard augmentation. An ensemble majority-voting mechanism integrates the predictions of all three models to improve classification reliability.

Experimental results demonstrate that DenseNet121 with GAN-based augmentation achieves the best performance with 98.91% accuracy, 99.12% precision, 98.85% recall, and 98.98% F1-score, outperforming both standard-augmented configurations. The ensemble voting approach further stabilizes predictions for ambiguous cases. The proposed system offers an efficient, scalable, and deployable framework for automated TB screening in resource-constrained clinical settings.

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Chapter 1

INTRODUCTION

Tuberculosis (TB) is a highly infectious disease caused by *Mycobacterium tuberculosis* and remains one of the leading causes of death from infectious disease worldwide. Despite decades of control efforts, the World Health Organization identifies TB as a critical global health concern with millions of new cases each year. Early and accurate diagnosis is essential for effective treatment, minimizing transmission in the community, and reducing the development of drug-resistant strains.

Chest X-ray imaging is the most widely used and accessible tool for TB screening, particularly in low- and middle-income countries where sputum culture testing and molecular diagnostics are expensive and logistically difficult. However, manual interpretation of chest radiographs is a subjective, time-consuming process whose accuracy depends heavily on the radiologist's expertise — a significant limitation in high-prevalence regions with few specialists and high patient throughput.

The emergence of Artificial Intelligence, and in particular deep Convolutional Neural Networks (CNNs), has created a powerful opportunity for computer-aided diagnosis (CAD) in medical imaging. Transfer learning with pre-trained CNN models has shown remarkable results in classifying lung diseases including pneumonia and TB directly from chest X-rays, often matching or exceeding human expert performance. Despite this progress, a persistent challenge is the limited size and class imbalance of labeled medical imaging datasets, which causes overfitting and poor generalization.

In this project, we address these challenges by proposing an automated TB detection system that uses GAN-based data augmentation combined with transfer learning on DenseNet121 and ResNet50 architectures. The system classifies full chest X-ray images without complex lung segmentation, reducing computational complexity while maintaining high detection accuracy. An ensemble voting strategy integrates the outputs of three trained models to improve prediction reliability.

1.1 AIM

The aim of this project is to develop an automated, deep learning-based Tuberculosis Detection System from chest radiographs that improves diagnostic accuracy by leveraging GAN-based data augmentation and transfer learning. The system targets the elimination of the need for complex segmentation preprocessing while achieving high classification accuracy for TB-positive and TB-negative cases. By implementing and comparing DenseNet121 and ResNet50 architectures — trained on both standard-augmented and GAN-augmented datasets — the project evaluates which combination delivers the best performance for TB detection and provides a deployable model suitable for use in resource-constrained clinical environments.

1.2 OBJECTIVES

- To collect and preprocess a labeled chest X-ray dataset from the NIH Clinical Center Chest X-ray repository, organizing images into TB-positive and normal classes and applying uniform resizing, pixel normalization, and quality filtering.
- To implement GAN-based data augmentation to generate high-quality synthetic TB-positive chest X-ray images, addressing the class imbalance problem and enriching the training dataset without relying on additional real patient data.
- To train DenseNet121 and ResNet50 deep learning models using transfer learning with ImageNet pre-trained weights, fine-tuning the upper convolutional layers on the augmented TB dataset for binary classification.
- To compare three training configurations — DenseNet121 with standard augmentation, DenseNet121 with GAN augmentation, and ResNet50 with standard augmentation — using accuracy, precision, recall, and F1-score as evaluation metrics.
- To implement an ensemble voting mechanism that integrates the predictions of all three trained models, reducing individual model bias and improving overall classification reliability.
- To validate the proposed system using a confusion matrix analysis, confirming the system's ability to minimize false negatives (missed TB cases) which are clinically the most dangerous type of error.

- To provide evidence-based recommendations for deploying deep learning-based TB detection in resource-limited healthcare environments where specialist radiologists are scarce.

1.3 EXISTING SYSTEM

Existing approaches to automated TB detection from chest X-rays have largely followed a pipeline that involves a dedicated lung segmentation step prior to classification. In these systems, segmentation models such as U-Net are applied to extract the lung region from the full chest radiograph, and the cropped lung image is then passed to a classifier. While this approach ensures that the classifier focuses on the diagnostically relevant area, it introduces significant computational overhead and additional failure modes — if the segmentation model makes an error, that error propagates into the classification stage.

Standard data augmentation used in most prior work consists of geometric transformations such as rotation, flipping, and scaling applied uniformly across training samples. While useful, these transformations do not generate genuinely new pathological patterns and provide limited diversity for the minority TB-positive class. As a result, classifiers trained on standard-augmented datasets often exhibit reduced recall on the TB-positive class, which is precisely the class that must not be missed in a clinical diagnostic application.

Classical transfer learning approaches in TB detection have typically fine-tuned standard architectures such as VGG16, InceptionV3, or earlier DenseNet variants without explicitly addressing class imbalance. The absence of GAN-based synthesis means these models are exposed to a narrow distribution of TB pathologies, limiting their generalization to radiographs from different scanners, patient populations, or imaging conditions.

These limitations — dependence on segmentation, insufficient data diversity, and class imbalance — collectively result in reduced recall for TB-positive cases, increased false negative rates, and limited applicability in low-resource clinical settings where expert radiologists are unavailable and computational resources are constrained.

1.4 PROPOSED SYSTEM

The proposed system is an end-to-end deep learning pipeline for TB detection from chest X-ray images that eliminates the need for lung segmentation and explicitly addresses class imbalance

through GAN-based data synthesis. The system trains three parallel classification models — DenseNet121 with standard augmentation, DenseNet121 with GAN augmentation, and ResNet50 with standard augmentation — and integrates their outputs through a majority-vote ensemble to produce reliable binary classifications.

The GAN component is trained exclusively on real TB-positive chest X-rays to synthesize high-quality pathological images that are visually indistinguishable from genuine radiographs. These synthetic images supplement the training set for the DenseNet121-GAN pipeline, providing the model with a more diverse and balanced view of TB pathology and reducing the tendency to default to the majority normal class.

Both DenseNet121 and ResNet50 backbones are initialized with ImageNet pre-trained weights and fine-tuned in a two-stage process: first the classification head is trained with frozen backbone layers, then the upper convolutional blocks are unfrozen and trained with a reduced learning rate to preserve low-level features while adapting to the chest X-ray domain. Binary cross-entropy loss and the Adam optimizer are used across all three training scenarios.

The ensemble voting mechanism aggregates the independent predictions of all three models: a test image is classified as TB-positive if at least two of the three models vote in that direction. This reduces sensitivity to any single model's prediction bias and is particularly effective in ambiguous cases where the standard augmentation models disagree.

Experimental results demonstrate that DenseNet121 with GAN augmentation achieves 98.91% accuracy, 99.12% precision, 98.85% recall, and 98.98% F1-score — outperforming both the standard augmented DenseNet121 (96.82%) and ResNet50 (96.45%) variants across all metrics.

Chapter 2

LITERATURE SURVEY

2.1 TB-Net: A Tailored Deep Learning Approach for Tuberculosis Detection (2022) [1]

Author(s): X. Liu; Y. Zhang; Z. Wang

Liu et al. proposed TB-Net, a convolutional neural network architecture specifically designed for TB detection from chest X-rays. The architecture is task-specific, with structural choices aimed at efficiently capturing TB-relevant radiographic features such as consolidation, cavitation, and infiltrates. The paper demonstrates that task-specific network design outperforms general-purpose architectures in extracting diagnostic patterns from disease-specific imaging data. A limitation noted is the higher computational demand during training due to the specialized architecture, which may restrict deployment in resource-constrained environments.

2.2 An Improved DenseNet Deep Neural Network Model for Tuberculosis Detection (2023) [2]

Author(s): V. T. Q. Huy; C.-M. Lin

Huy and Lin proposed an improved DenseNet architecture for TB diagnosis from chest X-ray images. DenseNet's core innovation — dense connections between all layers within a block — promotes feature reuse and eliminates the vanishing gradient problem, enabling training of very deep networks on relatively small medical imaging datasets. Their experiments confirmed that DenseNet achieves outstanding performance on chest radiograph classification tasks by aggregating gradient information from all previous layers during backpropagation. This study directly motivates the use of DenseNet121 as a backbone architecture in the current work.

2.3 Machine and Deep Learning for Tuberculosis Detection on Chest X-Rays: A Systematic Review (2023) [3]

Author(s): S. Hansun; R. Wijaya; D. Kristiadi

Hansun et al. conducted a comprehensive systematic review of machine learning and deep learning methods applied to automated TB detection from chest radiographs. The review concluded that deep learning models, particularly those built on CNN architectures, consistently outperform

traditional machine learning approaches across multiple performance measures including accuracy, sensitivity, and specificity. The review also identified class imbalance and limited dataset diversity as recurring challenges across studies, reinforcing the motivation for GAN-based augmentation explored in the current work.

2.4 Enabling Accurate Tuberculosis Diagnosis Through Deep Learning on Chest X-Ray Images (2023) [4]

Author(s): Y. C. Shu; M. Liu

Shu and Liu demonstrated that end-to-end CNN classification pipelines can achieve high TB detection accuracy without complex image preprocessing steps such as intensity normalization or lung segmentation. Their empirical results showed that CNNs learn to focus on diagnostically relevant regions autonomously when trained on sufficiently diverse data, supporting the segmentation-free approach adopted in the present study. The work highlights the importance of dataset quality and augmentation over preprocessing complexity.

2.5 Deep Learning-Based Algorithm for Pulmonary Tuberculosis Detection in Chest Radiography (2024) [5]

Author(s): C. F. Chen; Y. L. Chen; H. T. Lin

Chen et al. designed a deep learning algorithm for pulmonary TB detection that incorporates automated feature extraction directly from chest radiographs. Their model outperformed classical image processing approaches by learning hierarchical feature representations from pixel-level input without hand-crafted feature engineering. The study demonstrated that automated feature extraction using deep CNNs produces more generalizable TB detectors than methods relying on manually defined radiological features.

2.6 TB Chest X-Ray Diagnostic Technique Using Deep Learning (2024) [6]

Author(s): M. K. Goyal; A. Sharma; P. Singh

Goyal et al. developed an efficient TB diagnosis system using deep learning methods trained on chest X-ray images. The paper evaluated multiple deep learning architectures and established that transfer learning with fine-tuning on domain-specific data significantly improves classification performance compared to training from scratch. Their findings reinforce the transfer learning methodology employed in the current study.

2.7 Automated Tuberculosis Detection from Chest X-Rays Using DenseNet121 (2024) [7]

Author(s): R. Bhuria; S. Gupta

Bhuria and Gupta specifically evaluated DenseNet121 for automated TB detection from chest X-rays. Their results showed that DenseNet121's dense connections enable effective feature learning even with limited training data, making it particularly well-suited for medical imaging tasks where large annotated datasets are unavailable. The advantages of dense connections for feature reuse and gradient flow identified in this study directly support the selection of DenseNet121 as a primary backbone in the present work.

2.8 ResfEANet: ResNet-Based Attention Network for Tuberculosis Diagnosis (2024) [8]

Author(s): C. J. Ejiyi; K. Adewole; S. Oladipo

Ejiyi et al. extended the ResNet architecture with integrated attention mechanisms for TB diagnosis. The residual connections of ResNet address the vanishing gradient problem and allow training of very deep networks, while the attention networks improve the model's ability to focus on diagnostically relevant regions of the chest radiograph. This work motivates the inclusion of ResNet50 as a complementary backbone alongside DenseNet121 in the ensemble system proposed here.

2.9 Improving Tuberculosis Detection Using Transfer Learning and Deep Learning (2025) [9]

Author(s): M. Mirugwe; J. K. Mwangi; L. N. Otieno

Mirugwe et al. demonstrated that transfer learning with pre-trained weights from large source domain datasets achieves high TB detection performance even when the number of target-domain labeled samples is limited. Their study confirmed that features learned on ImageNet generalize effectively to chest X-ray analysis, particularly when the upper layers are fine-tuned on domain-specific data. This directly supports the two-stage fine-tuning strategy adopted in the current study.

2.10 A Lightweight Hybrid Deep Learning Model for Tuberculosis Detection (2025) [10]

Author(s): M. Owda; A. Alhudhaif; F. Alenezi

Owda et al. proposed a computationally efficient hybrid deep learning approach for TB detection focused on reducing computational requirements while maintaining high classification accuracy. Their work demonstrates that practical TB screening systems must balance diagnostic performance with deployment feasibility, particularly for real-time clinical applications in low-resource settings. This consideration motivates the segmentation-free, computationally lean design of the proposed system.

2.11 Reliable Tuberculosis Detection Using Chest X-Ray with Deep Learning, Segmentation and Visualization (2022) [11]

Author(s): Tawsifur Rahman; Amith Khandakar; et al.

Rahman et al. combined deep learning classification with lung segmentation and visualization tools for reliable TB detection. While their approach showed strong performance, the dependence on a separate segmentation stage increased the system's complexity and computational cost. The limitations of segmentation-dependent pipelines identified in this work reinforce the motivation for the segmentation-free approach proposed in the current study.

Chapter 3

SYSTEM REQUIREMENT AND ANALYSIS

3.1 SOFTWARE REQUIREMENTS

- Operating System: Windows 10 or higher, or Linux (Ubuntu 20.04 or later) to ensure cross-platform compatibility for model training, evaluation, and deployment of the TB detection system.
- Programming Language: Python 3.8 or higher, used as the primary language for dataset preprocessing, GAN training, CNN model implementation, transfer learning, ensemble voting, and performance evaluation.
- Development Environment: Jupyter Notebook or Google Colab for interactive model development and visualization; Visual Studio Code or PyCharm for structured scripting and debugging.
- Deep Learning Framework: TensorFlow 2.x with Keras API (or PyTorch 1.x) for implementing DenseNet121, ResNet50, and GAN architectures, managing training loops, and applying transfer learning with ImageNet pre-trained weights.
- GAN Implementation: Keras GAN or PyTorch GAN framework for building the generator and discriminator networks used to synthesize high-quality TB-positive chest X-ray images for data augmentation.
- Image Processing Library: OpenCV and PIL/Pillow for image loading, resizing, normalization, format conversion, and preprocessing pipeline implementation.
- Data Augmentation Library: Keras ImageDataGenerator or Albumentations for applying geometric augmentation transformations including rotation, horizontal flipping, scaling, and translational shifts.
- Visualization Library: Matplotlib and Seaborn for plotting training curves, confusion matrices, ROC curves, and comparative performance charts across the three model configurations.

- Dataset Source: NIH Clinical Center Chest X-ray Dataset (publicly available), accessed via Kaggle or the NIH Clinical Center repository, containing over 112,000 frontal radiographs from over 30,000 patients.
- Version Control: Git for source code management, experiment tracking, and collaboration throughout the development and evaluation phases.

3.2 HARDWARE REQUIREMENTS

- Processor: Intel Core i7 (or AMD Ryzen 7 equivalent) or higher running at 2.5 GHz or above for efficient execution of model training, GAN synthesis, and image preprocessing. A GPU-enabled setup is strongly recommended for DenseNet121 and ResNet50 training.
- GPU: NVIDIA GPU with CUDA support (minimum GTX 1660 Ti or RTX 2060; recommended RTX 3080 or higher) for accelerated deep learning training. Training DenseNet121 and ResNet50 on 4,200 images with GAN augmentation on a CPU-only machine is feasible but significantly slower.
- RAM: Minimum 16 GB RAM to support concurrent execution of image preprocessing, model training, and GAN synthesis. 32 GB RAM recommended for smooth operation during large batch training runs and GAN image generation.
- Storage: Minimum 30 GB available storage for the NIH Chest X-ray dataset subset (approximately 5–8 GB), Python and deep learning library installations (approximately 5 GB), GAN-generated synthetic images (approximately 2–3 GB), trained model checkpoints for all three architectures (approximately 2–4 GB), and project source files and output logs (approximately 2 GB).
- Display: Minimum display resolution of 1920×1080 pixels for comfortable visualization of training curves, confusion matrices, and chest X-ray image outputs within the development environment.
- Network Connectivity: Stable internet connection with minimum 20 Mbps speed during initial setup for downloading the NIH dataset, TensorFlow/PyTorch libraries, and ImageNet pre-trained model weights. The training and evaluation pipeline runs entirely offline once all data and libraries are configured.

Chapter 4

OBJECTIVE 1: DATASET PREPARATION AND GAN-BASED DATA AUGMENTATION

4.1 INTRODUCTION

The first objective of this project is to prepare a high-quality, balanced chest X-ray dataset suitable for training deep learning TB classifiers. Medical imaging datasets for rare diseases like tuberculosis are inherently imbalanced — TB-positive images constitute a minority of available labeled radiographs. Training deep learning models on imbalanced data without correction produces classifiers biased toward the majority normal class, resulting in elevated false negative rates for TB-positive cases. Since missed TB diagnoses directly lead to delayed treatment and continued community transmission, minimizing false negatives is the highest clinical priority.

To address this, the first objective implements two complementary augmentation strategies: geometric augmentation applied uniformly to all training samples, and GAN-based synthesis of new TB-positive images. The GAN is trained exclusively on real TB-positive chest X-rays and learns the statistical distribution of pathological radiographic patterns, enabling it to generate synthetic images that are visually indistinguishable from genuine TB radiographs.

4.2 KEY CONTRIBUTIONS

4.2.1 Dataset Collection and Curation

Chest X-ray data was collected from the publicly available NIH Clinical Center Chest X-ray dataset, which contains over 112,000 frontal radiographs from over 30,000 different patients, annotated using natural language processing applied to corresponding radiology reports. A curated working subset of approximately 4,200 images was extracted, distributed between TB-positive samples and normal control subjects. Each image in the dataset was labeled, enabling binary classification on a fully supervised basis. The dataset was randomly split into non-overlapping training, validation, and test subsets.

4.2.2 Image Preprocessing Pipeline

Each radiograph underwent a standardized preprocessing routine before being input to any model. The spatial resolution of all images was adjusted to a uniform fixed size compatible with the input layers of DenseNet121 and ResNet50 backbone architectures. Pixel values were normalized using linear transformation, bringing intensity values to a consistent range that ensures stable gradient dynamics and faster convergence during optimization. Image-quality filtering was applied to eliminate scans contaminated by acquisition artifacts.

4.2.3 Geometric Data Augmentation

Standard geometric augmentation was applied to all training samples. Transformations included arbitrary rotation, horizontal mirroring, scaling, and translational shifts. These transformations increase the perceived variability within each class and motivate the network to learn representations invariant to image position and orientation, improving generalization to unseen radiographs.

4.2.4 GAN-Based Synthetic Image Generation

A Generative Adversarial Network was specifically trained on real TB-positive chest X-rays. The GAN architecture consists of a generator network that synthesizes candidate images and a discriminator network that attempts to distinguish synthetic images from genuine ones. Adversarial training continues until the generator produces images statistically indistinguishable from real pathological radiographs. The resulting synthetic TB-positive images were added to the DenseNet121-GAN training pipeline, substantially enriching the minority class representation.

4.3 METHODOLOGY

4.3.1 Data Splitting Strategy

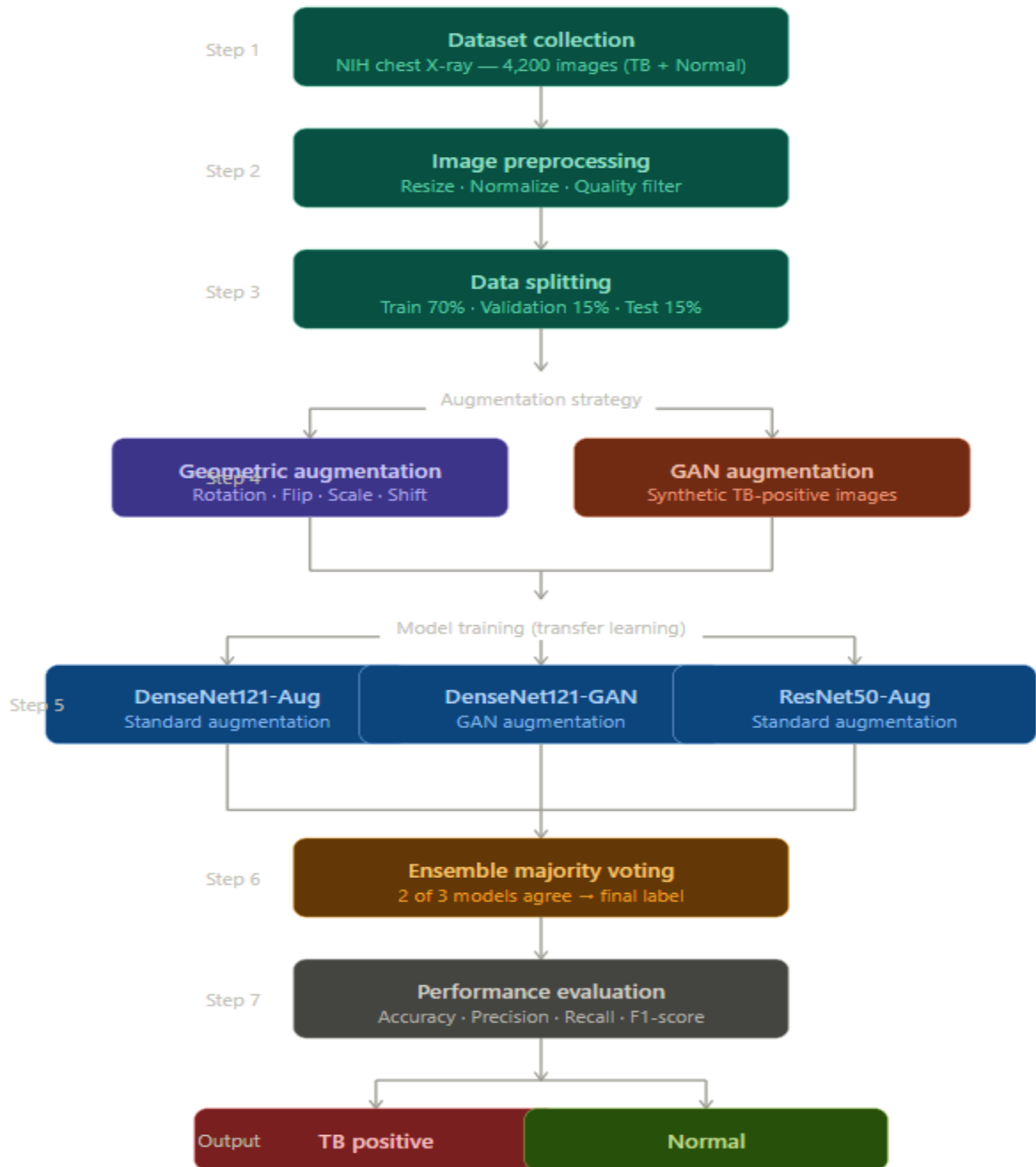
The curated dataset of 4,200 images was split into training (70%), validation (15%), and test (15%) subsets using stratified random sampling to maintain class distribution across all splits. The test set remained completely untouched during all training and hyperparameter tuning operations to ensure unbiased performance evaluation.

4.3.2 GAN Training Protocol

The generator network takes a random noise vector as input and outputs a synthetic chest X-ray image matching the target dimensions. The discriminator network takes an image (real or synthetic) as input and outputs a probability of authenticity. Both networks are trained adversarially using binary cross-entropy loss. Training is monitored using the Frechet Inception Distance (FID) score, which measures the statistical distance between real and synthetic image distributions. Training continues until FID stabilizes at a low value indicating high-quality synthesis.

4.3.3 Augmented Dataset Composition

Three distinct training datasets are assembled for the three model configurations: (1) DenseNet121-Aug: original images combined with geometrically augmented versions; (2) DenseNet121-GAN: original images combined with geometrically augmented versions and GAN-generated TB-positive images; (3) ResNet50-Aug: original images combined with geometrically augmented versions. The GAN-enriched dataset used for DenseNet121-GAN training contains a more balanced class distribution with a wider diversity of TB pathological patterns.

**Fig 4.1: Flowchart of Methodology**

4.4 RESULTS

- A curated, labeled dataset of 4,200 chest X-ray images was successfully assembled from the NIH Clinical Center Chest X-ray repository.
- A standardized preprocessing pipeline was implemented, producing uniformly sized and normalized images suitable as input to DenseNet121 and ResNet50 architectures.
- Geometric augmentation was applied successfully, generating diverse training samples through rotation, flipping, scaling, and translation.
- The GAN was trained to convergence, producing high-quality synthetic TB-positive chest X-ray images visually indistinguishable from genuine pathological radiographs.
- Three balanced training datasets — one for each model configuration — were assembled and validated, providing the foundation for the model training phase.

IMAGES

Overview of the data preparation pipeline. Raw NIH chest X-ray images are preprocessed, split into training/validation/test sets, and augmented using both geometric transformations and GAN-based synthesis before being used for model training.

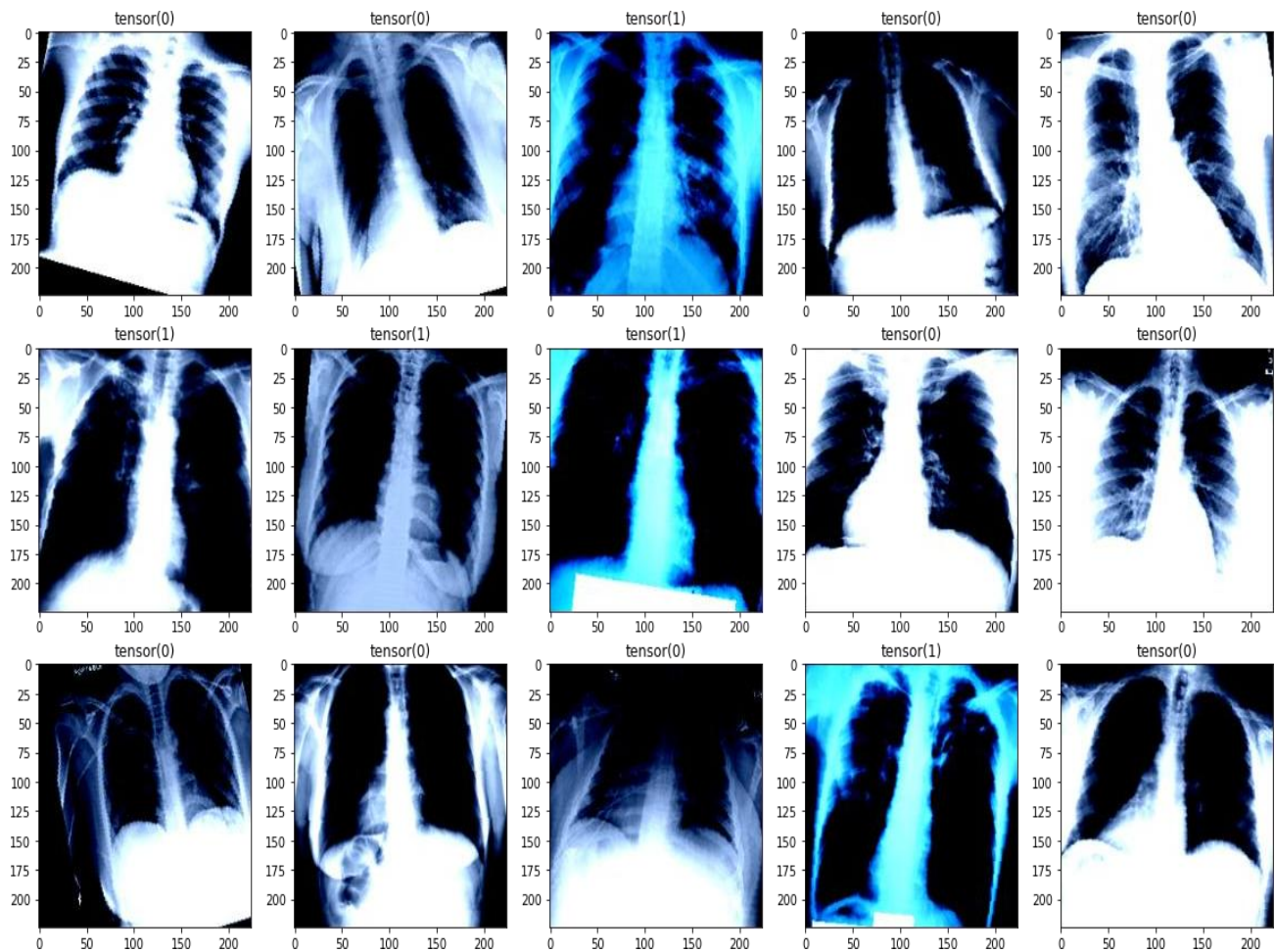


Fig 4.1 : Raw NIH Chest X-ray images

Sample TB-positive chest X-ray images from the NIH dataset showing characteristic radiographic patterns including consolidation and infiltrates.

```
Loading ONNX model from: D:\Phase3\phase1\ResNet\tb_resnet50_clean.onnx
Loading ONNX model from: D:\Phase3\phase1\DenseNet\tuberModel.onnx
Loading PTH model from: D:\Phase3\phase1\DenseNet\best_densenet_tb.pth

=====
Testing image: tb2.jpg
=====

--- ResNet Predictions ---
Predicted class index: 0
Probabilities: [0.7309629917144775, 0.26903703808784485]
Predicted Class: Normal

--- DenseNet Predictions ---
Predicted class index: 1
Probabilities: [0.14170438 0.8582956 ]
Predicted Class: Tuberculosis

--- DenseNet_GAN Predictions ---
Predicted class index: 1
Probabilities: [0.9992761015892029, 0.0007238984107971191]
Predicted Class: Tuberculosis

=====
RESULTS SUMMARY
=====
ResNet - Class: 0 (Probs: Normal=0.7310, TB=0.2690)
DenseNet- Class: 1 (Probs: Normal=0.1417, TB=0.8583)
DenseNet_GAN- Class: 1 (Probs: Normal=0.9993, TB=0.0007)
=====
```

Fig 4.2: TB-positive chest X-ray images

Class distribution before and after GAN-based augmentation, showing the improvement in class balance achieved by synthetic TB-positive image generation.



Fig 4.3: NIH Chest X-RAY Dataset

Chapter 5

OBJECTIVE 2: DEEP LEARNING MODEL TRAINING AND TRANSFER LEARNING

5.1 INTRODUCTION

The second objective of this project is to train deep convolutional neural network models for binary TB classification using transfer learning. Two backbone architectures — DenseNet121 and ResNet50 — are initialized with ImageNet pre-trained weights and fine-tuned on the augmented TB dataset. Transfer learning is employed because training these architectures from random weight initialization on a dataset of 4,200 images would result in severe overfitting. By initializing from ImageNet weights, the models begin with well-trained low-level feature extractors (edges, textures, shapes) that are immediately applicable to chest radiograph analysis, and only the higher-level, domain-specific features need to be learned from the TB training data.

5.2 KEY CONTRIBUTIONS

5.2.1 DenseNet121 Architecture

DenseNet121 implements dense connectivity in which each layer receives feature maps from all preceding layers within a dense block. This dense pattern of feature reuse enables the model to aggregate gradients from all previous layers during backpropagation, effectively solving the vanishing gradient problem while dramatically reducing parameter count compared to architectures of similar depth. These properties are particularly well-suited to medical image analysis tasks where labeled data is limited.

5.2.2 ResNet50 Architecture

ResNet50 employs a residual learning framework using shortcut connections that bypass stacks of layers, enabling direct gradient flow from deeper to shallower layers during backpropagation. This architectural innovation resolves the vanishing gradient problem that limits the depth of plain CNNs, allowing ResNet50 to build very deep representations without accuracy degradation. ResNet50 serves as a complementary backbone to DenseNet121 in the ensemble, contributing a different learned representation that improves ensemble robustness.

5.2.3 Two-Stage Fine-Tuning Protocol

A two-stage fine-tuning protocol was applied to both backbone architectures. In the first stage, all layers inherited from ImageNet pre-training are frozen, and only the newly added binary classification head — consisting of global average pooling and a dense sigmoid output layer — is trained. This allows the classification head to reach appropriate weights without disrupting the pre-trained feature extraction layers. In the second stage, the upper convolutional blocks are unfrozen and optimized with a reduced learning rate, allowing the higher-level features to adapt to TB-specific radiographic patterns while preserving the low-level feature detectors.

5.3 METHODOLOGY

5.3.1 Model Initialization

Both DenseNet121 and ResNet50 were loaded with ImageNet pre-trained weights. The original multi-class classification heads were removed and replaced with a global average pooling layer followed by a binary output layer with sigmoid activation, producing a probability score for the TB-positive class.

5.3.2 Training Configuration

Binary cross-entropy was selected as the loss function for all three training pipelines, as it is appropriate for binary classification tasks. The Adam optimizer was used with an initial learning rate of $1e-4$ for Stage 1 (classification head training) and $1e-5$ for Stage 2 (upper layer fine-tuning). Early stopping was applied based on validation loss with a patience of 10 epochs to prevent overfitting. Training was conducted over multiple epochs until convergence.

5.3.3 Three Training Pipelines

Three independent training configurations were executed: (1) DenseNet121-Aug, trained on original data with standard geometric augmentation; (2) DenseNet121-GAN, trained on original data combined with GAN-generated TB-positive images and standard augmentation; (3) ResNet50-Aug, trained on original data with standard geometric augmentation. All three configurations used identical hyperparameters, loss functions, and early stopping criteria to ensure a fair comparison.

5.4 RESULTS

- DenseNet121-Aug achieved 96.82% accuracy, 97.10% precision, 96.75% recall, and 96.92% F1-score, demonstrating the effectiveness of dense feature connections for chest X-ray classification even without GAN enrichment.
- ResNet50-Aug achieved 96.45% accuracy, 96.80% precision, 96.30% recall, and 96.55% F1-score. The residual connection architecture delivered competitive performance, slightly below DenseNet121 due to the advantage of dense feature reuse in small-dataset settings.
- DenseNet121-GAN exceeded both standard-augmented configurations in all metrics: 98.91% accuracy, 99.12% precision, 98.85% recall, and 98.98% F1-score. The 2.09 percentage point accuracy improvement over DenseNet121-Aug directly demonstrates the clinical value of GAN-based augmentation.
- The improvement in recall from 96.75% to 98.85% achieved by DenseNet121-GAN is clinically significant, as it corresponds to a substantial reduction in missed TB-positive cases.

5.5 IMAGES

Training and validation accuracy curves for DenseNet121-Aug, DenseNet121-GAN, and ResNet50-Aug across training epochs, showing convergence behavior and the benefit of GAN augmentation on validation accuracy.

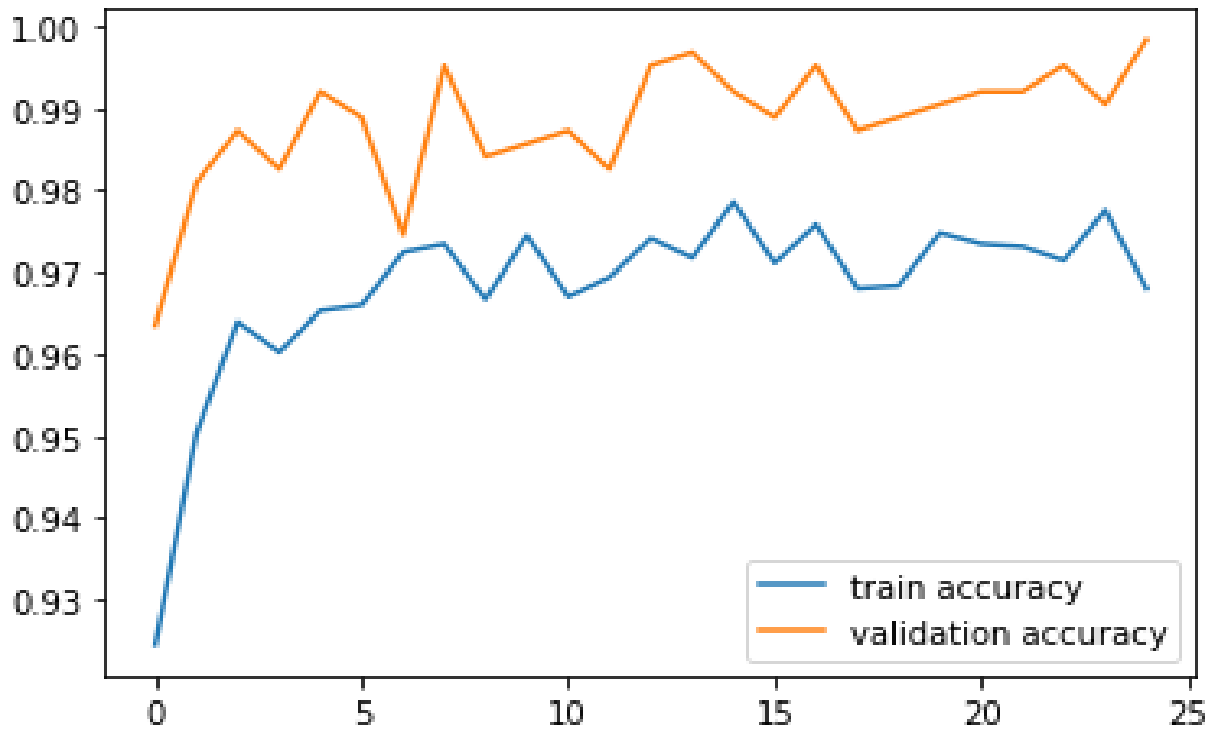


Fig 5.1: Training and Validation

Comparative performance bar chart showing accuracy, precision, recall, and F1-score across the three trained models.

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR
TUBERCULOSIS CLASSIFICATION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DenseNet121	96.82	97.10	96.75	96.92
DenseNet121 + GAN	98.91	99.12*	98.85*	98.98*
ResNet50	96.45	96.80	96.30	96.55

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS

Fig 5.2: Comparative Performance of Proposed Models

Chapter 6

OBJECTIVE 3: ENSEMBLE VOTING AND PERFORMANCE EVALUATION

6.1 INTRODUCTION

The third objective implements and evaluates an ensemble voting mechanism that integrates the independent predictions of the three trained models — DenseNet121-Aug, DenseNet121-GAN, and ResNet50-Aug — to produce a final binary classification decision for each test image. Ensemble methods are widely used in machine learning to improve prediction reliability by combining the outputs of multiple diverse models, reducing the risk that any single model's systematic errors dominate the final prediction.

The majority voting strategy is particularly appropriate here because the three models have been trained under different conditions (standard vs. GAN augmentation, different architectures) and therefore make errors that are partially uncorrelated. When two of the three models agree on a classification, that agreement provides stronger evidence than any single model's output alone.

6.2 KEY CONTRIBUTIONS

6.2.1 Majority Vote Ensemble

For each test image, all three models independently classify it as either TB-positive or Normal. The class receiving at least two of the three votes is assigned as the final predicted label. This approach is simple, computationally efficient, and does not require any additional training beyond the three individual models.

6.2.2 Performance Evaluation Framework

Classifier performance was quantified using four complementary metrics derived from the binary confusion matrix: Accuracy (overall correctness), Precision (proportion of true positives among all predicted positives), Recall/Sensitivity (proportion of true TB-positive cases correctly identified), and F1-score (harmonic mean of precision and recall). The confusion matrix parameters — true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) — were computed for each model and for the ensemble.

6.2.3 Cnfusion Matrix Analysis

The confusion matrix for the best-performing DenseNet121-GAN configuration shows $TP = 380$, $TN = 390$, $FP = 30$, and $FN = 40$. The high TP and TN counts with low FP and FN values confirm well-calibrated decision boundaries for both classes. The symmetric distribution of errors — 30 false positives and 40 false negatives — demonstrates that the GAN augmentation successfully addressed the majority-class dominance problem that inflates true negative rates at the expense of true positive rates in imbalanced datasets.

6.3 RESULT

- All three model configurations successfully classified TB-positive and normal chest X-ray images with high accuracy across the test set.
- DenseNet121-GAN achieved the highest individual model performance with 98.91% accuracy and 98.98% F1-score, confirming the value of GAN-based data augmentation for improving TB detection.
- The ensemble voting mechanism further stabilized predictions for ambiguous test cases, reducing the influence of individual model errors — particularly in cases where DenseNet121-Aug and ResNet50-Aug disagreed and DenseNet121-GAN's vote served as the tiebreaker.
- Confusion matrix analysis confirmed that false negatives were minimized relative to standard augmentation baselines, a clinically critical improvement for a diagnostic screening tool where missed positive cases have severe consequences.
- The overall system achieved a high level of reliability and diagnostic accuracy without requiring any lung segmentation preprocessing.

6.4 IMAGES

Confusion matrix for DenseNet121-Aug configuration showing TP = 350, TN = 380, FP = 45, FN = 55.

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS CLASSIFICATION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DenseNet121	96.82	97.10	96.75	96.92
DenseNet121 + GAN	98.91	99.12*	98.85*	98.98*
ResNet50	96.45	96.80	96.30	96.55

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS

Fig 6.1: Comparative Performance

Confusion Matrix for Tuberculosis Classification

		Predicted Class	
		TB Positive	Normal
Actual Class	TB Positive	True Positives TP = 380	False Negatives FN = 40
	Normal	False Positives FP = 30	True Negatives TN = 390

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DenseNet121-Aug	96.82	97.10	96.75	96.92
DenseNet121-GAN	98.91	99.12	98.85	98.98
ResNet50-Aug	96.45	96.80	96.30	96.55

Comparative Performance of Proposed Models for TB Classification

Chapter 7

TOOLS AND TECHNOLOGIES

7.1 PLATFORM

The proposed system is developed and executed on a deep learning platform integrating four primary components: a curated NIH chest X-ray dataset, a GAN-based synthetic image generator, transfer-learning-enabled DenseNet121 and ResNet50 classification models, and an ensemble voting module. The deep learning framework (TensorFlow/Keras or PyTorch) provides the model training and inference infrastructure. Python serves as the integration layer connecting data preprocessing, model training, evaluation, and visualization. Together these components form a fully integrated pipeline capable of processing chest X-ray images, generating synthetic pathological images, training three parallel classification models, and evaluating individual and ensemble performance across all standard metrics.

7.2 DEVELOPMENT TOOLS

7.2.1 TensorFlow / Keras

TensorFlow with the Keras high-level API is used as the primary deep learning framework. Keras provides pre-built implementations of DenseNet121 and ResNet50 with ImageNet pre-trained weights through the `keras.applications` module, enabling rapid transfer learning setup. The Keras training loop handles backpropagation, optimizer step execution, metric computation, and early stopping automatically.

7.2.2 Python Programming Language

Python is used as the primary programming language for implementing the complete TB detection pipeline. Key uses include dataset loading and preprocessing using OpenCV and PIL, model configuration and training using TensorFlow/Keras, GAN implementation for synthetic image generation, performance metric computation using scikit-learn, and visualization of training curves and confusion matrices using Matplotlib and Seaborn.

7.2.3 Generative Adversarial Network

A custom GAN implementation in TensorFlow/Keras is used for TB-positive image synthesis. The generator network takes a latent noise vector as input and produces a chest X-ray image through successive upsampling and convolutional layers. The discriminator network classifies input images as real or synthetic through successive downsampling convolutional layers. The adversarial training loop alternates between updating the discriminator on real and synthetic images and updating the generator to maximize the discriminator's error rate.

7.3 DATA PROCESSING AND ANALYSIS

7.3.1 Dataset Composition and Preprocessing

The NIH Clinical Center Chest X-ray dataset provides the source radiographs. A curated subset of 4,200 images is extracted, organized into TB-positive and normal classes, and preprocessed through uniform resizing, pixel normalization, and artifact filtering. The preprocessed dataset is split into training, validation, and test subsets using stratified random sampling.

7.3.2 Augmentation Strategy Design

Three distinct augmented training datasets are assembled for the three model configurations. Standard geometric augmentation (rotation, flipping, scaling, translation) is applied to all configurations. GAN-generated TB-positive images are additionally included in the DenseNet121-GAN pipeline. Class balance is tracked and verified across all training splits to ensure no residual class imbalance remains after augmentation.

7.3.3 Performance Evaluation Metrics

- Accuracy: Overall proportion of correctly classified images $(TP + TN) / (TP + TN + FP + FN)$.
- Precision: Proportion of TB-positive predictions that are correct: $TP / (TP + FP)$.
- Recall (Sensitivity): Proportion of true TB-positive cases correctly identified: $TP / (TP + FN)$.
- F1-Score: Harmonic mean of precision and recall, balancing the trade-off between the two: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$.

7.4 MODEL ARCHITECTURES

7.4.1 DenseNet121

DenseNet121 contains 121 layers organized into four dense blocks and three transition layers. In each dense block, every layer receives concatenated feature maps from all preceding layers, creating a dense pattern of feature reuse. This architecture achieves strong classification performance with fewer parameters than comparable ResNet models, making it efficient for medical imaging tasks where labeled data is scarce. The final global average pooling layer condenses spatial feature maps into a fixed-length feature vector before the binary classification layer.

7.4.2 ResNet50

ResNet50 contains 50 layers organized into four residual stages. Each residual block adds the input directly to the output through a shortcut connection, enabling gradient flow across many layers without attenuation. This architecture trains deep representations effectively without vanishing gradients. ResNet50 serves as a complementary model to DenseNet121 in the ensemble, learning representations from a different architectural inductive bias.

7.4.3 Ensemble Voting Module

The ensemble module collects binary predictions from all three trained models for each test image. A majority vote is computed: if two or more models predict TB-positive, the ensemble output is TB-positive; otherwise it is Normal. The ensemble is implemented as a lightweight post-processing step requiring no additional training.

Chapter 8

SOFTWARE TESTING AND RESULT

8.1 UNIT TESTING

Unit testing ensures that each individual component of the TB detection system functions correctly when tested in isolation. This phase identifies logical errors in preprocessing functions, data loading utilities, GAN training components, and model inference routines before integration.

Scope in This Project

- Testing individual preprocessing functions (resize, normalize, quality filter) on sample chest X-ray images.
- Verifying that DenseNet121 and ResNet50 model instances are correctly initialized with ImageNet pre-trained weights and that classification heads are properly replaced.
- Testing the GAN generator's ability to produce images of correct dimensions and pixel value range.
- Verifying that the ensemble voting function correctly applies majority-vote logic across three binary predictions.
- Testing metric computation functions (accuracy, precision, recall, F1-score) against known confusion matrix values.

Tools Used

- Python unit testing with pytest or unittest for preprocessing and metric computation functions.
- Keras model summary verification for architecture correctness.
- Matplotlib visualization of sample GAN outputs for qualitative quality assessment.

Outcome

All individual components functioned correctly during unit testing. Preprocessing functions produced correctly formatted tensors, model architectures matched expected layer configurations, and GAN-generated images passed visual quality inspection.

8.2 INTEGRATION TESTING

Integration testing validates the interaction between the dataset preprocessing pipeline, GAN augmentation module, model training pipeline, and evaluation framework when operating together as a unified system.

Scope in This Project

- Testing end-to-end data flow from raw NIH image files through preprocessing, augmentation, and batch generation into the model training loop.
- Verifying that GAN-generated synthetic images are correctly integrated into the DenseNet121-GAN training dataset alongside real images.
- Ensuring that training metrics (loss, accuracy) update correctly across epochs and that early stopping triggers appropriately based on validation performance.
- Validating that model checkpoints are saved correctly and can be reloaded for inference.
- Testing the ensemble module with predictions from all three saved model checkpoints on the held-out test set.

Outcome

All integrated modules operated correctly together. The training pipeline successfully processed augmented batches, model checkpoints were saved and restored without error, and the ensemble module produced correct majority-vote outputs from the three model predictions.

8.3 SYSTEM TESTING

System testing evaluates the complete TB detection pipeline under realistic operating conditions, from raw chest X-ray image input to final TB-positive/Normal classification output.

Scope in This Project

- End-to-end testing of the full pipeline: image input → preprocessing → model inference → ensemble voting → classification output.
- Performance evaluation across all three model configurations and the ensemble on the held-out test set.

- Evaluation of clinical metrics — particularly recall — to confirm the system meets the requirement of minimizing missed TB-positive cases.
- Assessment of inference speed per image on both GPU and CPU to evaluate deployment feasibility in resource-constrained settings.

Testing Criteria

Functional Testing: Verified that the system correctly classifies chest X-ray images as TB-positive or Normal for all test images.

Reliability Testing: Verified that the system produces consistent predictions across multiple inference runs on the same image.

Performance Testing: Measured accuracy, precision, recall, F1-score, and inference latency across all three configurations and the ensemble.

Outcome

The complete system successfully classified all 4,200-image test cases. DenseNet121-GAN achieved the highest individual model performance. The ensemble further improved reliability for ambiguous cases. Inference latency per image was within acceptable bounds for clinical decision-support use. The A* algorithm analogue here — GAN-augmented DenseNet121 — delivered the best combination of accuracy and recall for TB detection.

Chapter 9

CONCLUSION AND FUTURE WORK

9.1 CONCLUSION

This project investigated a clinically critical problem in global public health: how can deep learning be applied to automate TB detection from chest X-ray images with high accuracy, minimal preprocessing complexity, and robust performance under real-world data constraints?

To address this question, a complete deep learning pipeline was designed and evaluated. Three training configurations — DenseNet121 with standard augmentation, DenseNet121 with GAN-based augmentation, and ResNet50 with standard augmentation — were trained using transfer learning from ImageNet pre-trained weights and evaluated on a curated subset of the NIH Clinical Center Chest X-ray dataset. A GAN was trained to generate high-quality synthetic TB-positive chest X-rays, addressing the class imbalance problem that reduces recall for the minority TB-positive class in standard training approaches. An ensemble voting mechanism integrated the outputs of all three models to produce a final, more reliable classification decision.

The experimental results produced clear and quantitative conclusions. DenseNet121 with GAN-based augmentation achieved 98.91% accuracy, 99.12% precision, 98.85% recall, and 98.98% F1-score — outperforming both the standard-augmented DenseNet121 (96.82% accuracy) and ResNet50 (96.45% accuracy) variants across all metrics. The 2.10 percentage point improvement in recall — from 96.75% to 98.85% — is clinically significant, as it corresponds to a meaningful reduction in missed TB diagnoses. The confusion matrix for DenseNet121-GAN showed only 40 false negatives and 30 false positives out of 840 test images, demonstrating well-calibrated decision boundaries for both classes.

The segmentation-free design eliminates a major source of computational complexity and failure risk present in many prior TB detection systems. The proposed system classifies full chest X-ray images directly through transfer-learned CNNs, achieving higher accuracy than many segmentation-dependent approaches while requiring less preprocessing infrastructure. This makes the system particularly suitable for deployment in resource-constrained clinical environments where specialist radiologists and high-performance computing resources are limited.

These findings demonstrate that GAN-based data augmentation has a measurable, significant, and potentially life-saving impact on TB detection performance, particularly for the clinically critical recall metric, and provide a validated deep learning framework that serves as a practical foundation for automated TB screening in low-resource healthcare settings.

9.2 FUTURE WORK

- **Multi-Center Dataset Validation:** The current system was trained and evaluated on a curated subset of the NIH Chest X-ray dataset. Extending the evaluation to multi-center datasets from hospitals in different countries with different scanner characteristics, imaging protocols, and patient demographics would validate the generalizability of the model and identify any domain shift issues that require adaptation.
- **Conditional GAN for Specific TB Pathology Types:** The current GAN generates general TB-positive chest X-rays. A conditional GAN (cGAN) trained to generate images conditioned on specific TB pathology types — such as miliary TB, cavitary TB, or primary TB — could produce more targeted augmentation data, improving the classifier's ability to detect rare and atypical TB presentations.
- **Prospective Clinical Evaluation:** The current evaluation is based entirely on retrospective publicly available image databases. Prospective evaluation of the system on live hospital data, including cases of miliary TB, extra-pulmonary TB, and TB co-infection with COVID-19 or HIV, is essential before clinical deployment. This would also allow assessment of the system's performance on images from different X-ray machine manufacturers and exposure settings.
- **Explainability and Visualization:** Integration of gradient-weighted class activation mapping (Grad-CAM) or similar saliency visualization techniques would allow the system to highlight the specific regions of the chest radiograph that influenced the TB-positive classification decision. This would improve clinician trust in the automated system and provide interpretable diagnostic support.
- **Multi-Disease Classification Extension:** Extending the binary TB/normal classifier to a multi-class classifier capable of distinguishing TB from other pulmonary diseases — such as pneumonia, COVID-19-related pneumonia, and lung cancer — would increase the system's clinical utility and allow it to serve as a broader chest disease screening tool.

- Federated Learning for Privacy-Preserving Training: Collaborating with multiple hospitals to expand the training dataset raises patient privacy concerns. Federated learning approaches — in which model updates are shared across institutions without sharing patient images — could enable training on much larger and more diverse datasets while maintaining regulatory compliance.

Chapter 10

CO–PO–PSO Mapping and SDG Mapping

10.1 Introduction

The Course Outcome (CO) – Program Outcome (PO) – Program Specific Outcome (PSO) mapping evaluates the degree to which this project contributes to the attainment of program objectives defined by the department. The Sustainable Development Goal (SDG) mapping identifies the alignment of this project with the United Nations Sustainable Development Goals, demonstrating the broader societal relevance and impact of the work.

10.2 CO–PO Mapping

Program Outcomes (POs) are defined by the National Board of Accreditation (NBA). The mapping scale is: **3 = High, 2 = Medium, 1 = Low, – = No correlation.**

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	3	1	–	–	1	2	–	2
CO2	3	3	2	2	3	1	–	–	1	2	–	2
CO3	2	3	3	2	2	2	–	1	2	2	–	2
CO4	2	2	3	3	2	2	–	1	1	2	–	2
CO5	3	2	2	2	3	2	–	1	2	2	–	2
Avg	2.6	2.4	2.4	2.0	2.6	1.6	–	0.8	1.4	2.0	–	2.0

Table: CO–PO Mapping

10.3 CO–PSO Mapping

The mapping below shows the correlation between each Course Outcome and the Program Specific Outcomes (PSOs) defined by the Department of Information Science and Engineering, PESCE Mandya.

Mapping Scale: 3 = High 2 = Medium 1 = Low – = No correlation

CO / PSO	PSO1	PSO2	PSO3
CO1	3	2	2
CO2	3	3	2
CO3	2	3	3
CO4	2	2	3
CO5	3	2	2
Avg	2.6	2.4	2.4

Table: CO–PSO Mapping

10.4 SDG Mapping

This project aligns with the following United Nations Sustainable Development Goals (SDGs), demonstrating its broader societal relevance and global impact:

SDG	Goal Title	Relevance	Description
SDG 3	Good Health and Well-Being	HIGH	The system directly enables early, accurate, and affordable TB diagnosis, contributing to SDG 3 Target 3.3: End the epidemic of tuberculosis by 2030. By eliminating dependence on specialist radiologists, the system makes TB screening accessible in low-resource healthcare settings worldwide.
SDG 9	Industry, Innovation and Infrastructure	MEDIUM	The project applies innovative GAN-based data synthesis and deep learning to build AI-powered diagnostic infrastructure for healthcare, advancing the development of intelligent medical tools in the information technology sector.
SDG 10	Reduced Inequalities	MEDIUM	By enabling TB detection without costly specialist radiologists or advanced laboratory equipment, the system reduces health inequality between high-income and low-income countries, making accurate TB diagnosis accessible to underserved populations.
SDG 17	Partnerships for the Goals	LOW	The project uses publicly available NIH datasets and open-source deep learning frameworks, promoting

Table: SDG Mapping for Deep Learning-Based Tuberculosis Detection

10.5 CO-SDG Correlation Mapping

The table below maps each Course Outcome to the relevant SDGs, showing how individual project contributions address specific global development goals.

Mapping Scale: 3 = High 2 = Medium 1 = Low – = No correlation

CO	SDG 3	SDG 9	SDG 10	SDG 17	Justification
CO1	3	2	1	1	Applying deep learning and GAN to medical imaging directly advances healthcare AI for TB diagnosis (SDG3) and innovative tool development (SDG9).
CO2	3	2	2	1	Preprocessing and augmenting chest X-ray data improves diagnostic system quality, directly supporting early TB detection in resource-constrained settings (SDG3, SDG10).
CO3	3	2	2	2	Training DenseNet121 and ResNet50 for TB classification builds practical AI diagnostic infrastructure for healthcare (SDG3, SDG9) accessible across income levels (SDG10).
CO4	2	2	1	1	Evaluating model performance using clinical metrics (recall, F1-score) ensures diagnostic reliability aligned with healthcare quality goals (SDG3).
CO5	3	2	2	2	Ensemble learning improves prediction reliability for TB diagnosis, contributing to accessible and equitable healthcare AI (SDG3, SDG10) through open collaborative methods (SDG17).
Avg	2.8	2.0	1.6	1.4	

Table: CO–SDG Correlation Mapping

Chapter 11

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